Functional Factor Models for Intraday FX Volatility: Stability of the Loading Structure

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Introduction

Intraday FX volatility has shape

- Largest, most liquid market - trades 24h (Sydney, Tokyo, London, New York)
- Volatility peaks at session opens and the London–NY overlap
- A single daily realized-volatility number hides this intraday profile
- **Idea**: treat each day as a functional observation: a curve over $\tau \in [0, 1]$

The curves co-move

- Pairs react jointly to global events (macro releases, policy, geopolitical stress)
- Almost all cross-pair correlations positive $\rightarrow$ a market-wide volatility factor
- Two blocks: tightly-linked G10 vs. weak emerging-market cluster; USD/TRY isolated
- High-dimensional curves $\rightarrow$ summarise co-movement with a few common components
- Motivates a **Functional Factor Model** (Hallin, Nisol & Tavakoli, 2023)

Research Question

Is the loading subspace stable over time?

Apply

- Fit the FFM to 44 FX pairs over 2014–2025

Assess

- A loading is a *curve* $\rightarrow$ it can change two ways: *level* (amplitude) vs. *structure* (timing/shape)

Evaluate

- If the subspace rotates, the static model is misspecified $\rightarrow$ what does it cost forecasting?

A test on the mean loading would miss a pure rotation $\Rightarrow$ treat stability as a geometric question.

Outline

1. Introduction
2. Data & Volatility proxy
3. Functional Factor Model
4. Loading Structure Stability
5. Out-of-sample Performance
6. Conclusion

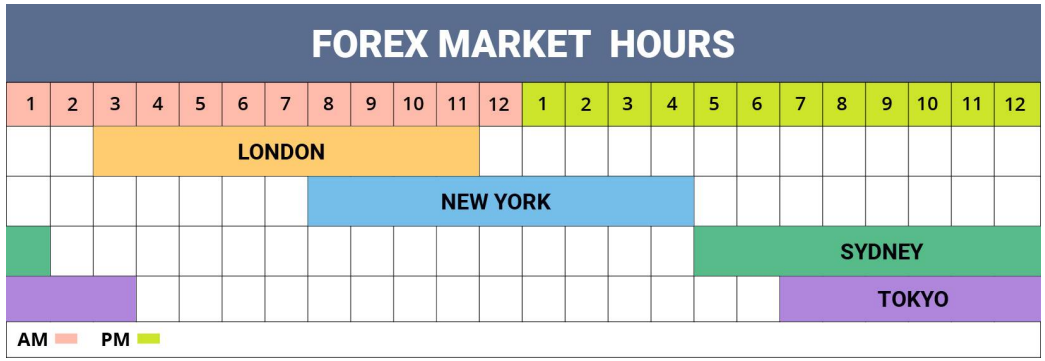
Data and pair selection

- HistData (MetaTrader) 1-min $\rightarrow$ 5-min (last tick); + Dukascopy USD/CNH
- Trading day 17:00 $\rightarrow$ 16:55 NY $\Rightarrow M = 288$ intraday points; weekends & Jan 1 removed
- 48 candidates but we retain only the trading days on which all pairs have at least 70% of their intraday observations available $\rightarrow$ drop EUR/HUF, EUR/TRY
- EUR/DKK, USD/HKD (pegged, ≈ 0 vol)

$\Rightarrow$ $N = 44$ **pairs,** $T = 3062$ **days**

Event	**NY time**	τ
Sydney open / NY close	17:00	0.00
Tokyo open	19:00	0.08
London open	03:00	0.42
New York open	08:00	0.62
London close	12:00	0.79
New York close	16:55	1.00

Table: Session boundaries mapped to τ.



Volatility proxy

- 5-min log-return:
  $r_{it}(\alpha) = \log P_{it}(\alpha) - \log P_{it}(\alpha - 1)$
  $\alpha =$ 5-minute timestamps
- Natural proxy $|r_{it}|$, but right-skewed $\rightarrow$ crisis points dominate the covariance & distort its eigenfunctions
- $\Rightarrow$ log-proxy $\log(r_{it}^2 + \epsilon_i)$
  ($\epsilon_i =$ 5th pct. of positive r_{it}^2, avoids $\log 0$)
- SV model: $\log(r_{it}^2) = \log(\sigma_{it}^2) + \log(\nu_{it}^2)$
  $\rightarrow$ additive signal + noise
- Noise: mean (≈ -1.27) absorbed in the curve mean; variance $\pi^2/2 \approx 4.93$ (large)

$\Rightarrow$ data still very noisy

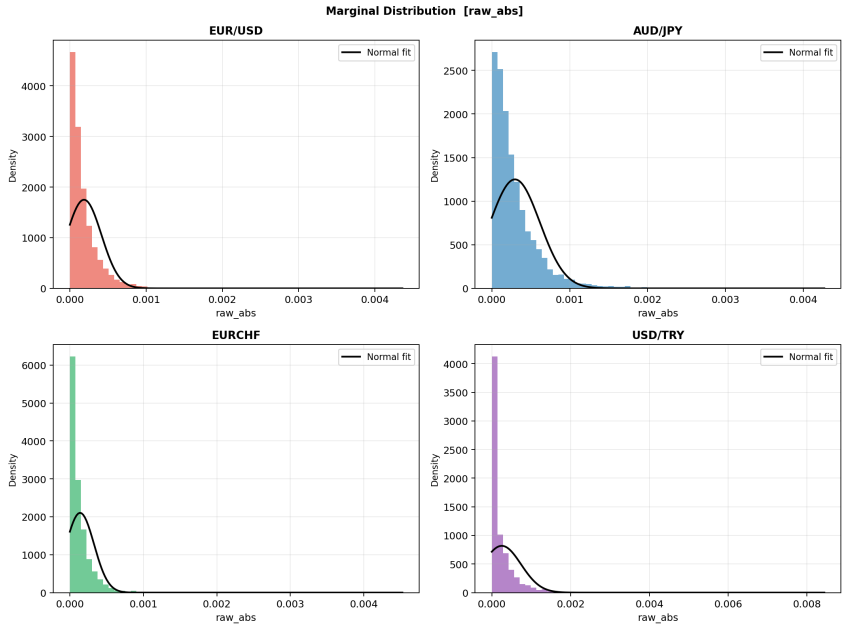


Figure: Raw $|r|$ vs. smoothed log-proxy, four pairs.

Penalized B-spline smoothing

- Each day = 288 noisy points → continuous curve in $L^2([0,1])$
- Cubic B-splines ($K = 20$) + 2nd-order difference penalty (P-splines)
- One λ_i per pair via GCV; liquid G10 smoothed most, spiky EM pairs more flexible
- **Robustness**: full pipeline re-run on raw proxy (App. E) - same factor & subspace

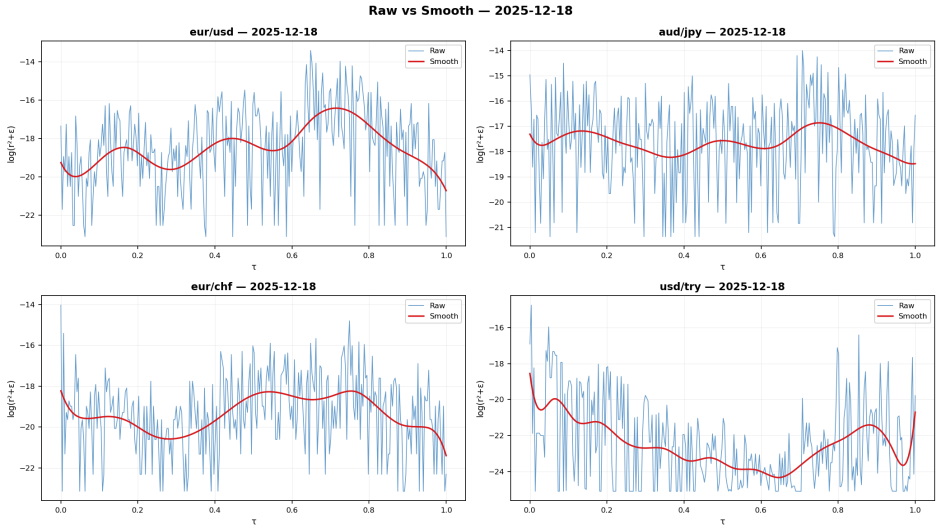


Figure: Raw vs. P-spline-smoothed curve, one day.

Exploratory Analysis

- Mean curves show clear session structure: U-shape + open bumps
- EUR/USD tight & regular; AUD/JPY active in all 3 sessions
- USD/TRY most dispersed (several regimes)

$\Rightarrow$ recurring cross-day shape justifies functional data

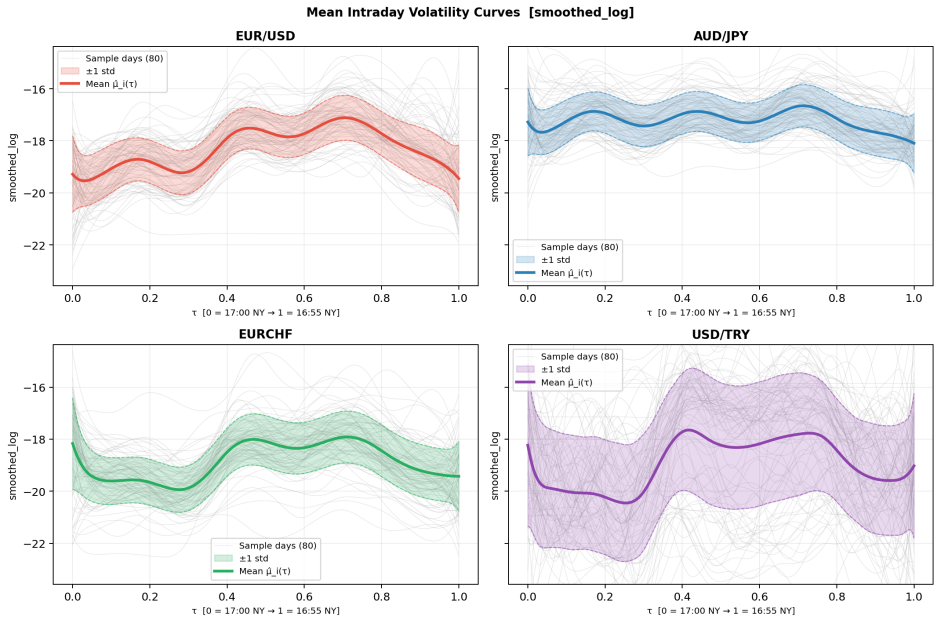


Figure: Mean intraday curve $\hat{\mu}_i(\tau)$ with ± 1 s.d. band.

Functional Factor Model

$$x_{it}(\tau) = \mu_i(\tau) + \sum_{k=1}^{r} b_{ik}(\tau) \cdot u_{kt} + \xi_{it}(\tau)$$

$i = 1, \ldots, N$ *pairs* $\quad t = 1, \ldots, T$ *days* $\quad \tau \in [0, 1]$ *(M intraday points)*

Notation

- $x_{it}(\tau)$: log-vol curve for pair i, day t, minute τ
- μ_i : mean intraday curve of pair i (pair-specific)
- $b_{ik}(\tau)$: functional loading - sensitivity of pair i to factor k at minute τ
- u_{kt} : scalar factor (common shock on day t)
- $\xi_{it}(\tau)$: idiosyncratic component (pair-specific residual)

Estimation (FPCA, Hallin et al.)

1. Gram matrix $F \in \mathbb{R}^{T \times T}$ (inner products between days)
2. Leading eigenpairs $\rightarrow$ scores $\hat{u}$ (unit variance)
3. Functional directions $\hat{e}_k$ (project days on scores)
4. Loadings $\hat{b}_{ik}(\tau)$ per pair
5. Recover $\hat{\chi}_{it}$ (common) & $\hat{\xi}_{it}$ (idiosyncratic)

Identified only up to rotation $\Rightarrow$ study the *subspace*

FFM estimation - Number of factor selection

Goal: select the unknown number of factors r.
Three complementary criteria:

- **IC** (Bai & Ng, functional): over-penalizes when $N \ll T \rightarrow$ always $\hat{r} = 1$
- **ABC** (Alessi et al.): median $\hat{r} = 2$, but unstable across permutations
- **ER / GR** (Ahn & Horenstein): penalty-free, peak sharply at $k = 1$; ER recommended for $N \ll T$

$\Rightarrow$ Adopt $\hat{r} = 1$ (ER & GR agree).

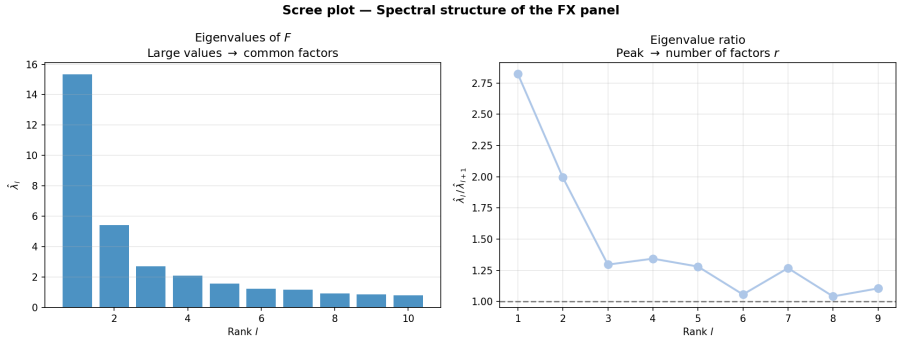


Figure: Eigenvalues of F and ratios $\hat{\lambda}_k / \hat{\lambda}_{k+1}$.

$\hat{\lambda}_1$: 27.8% of variance; ER peaks at 2.83 ($k = 1$).

FFM estimation - Results

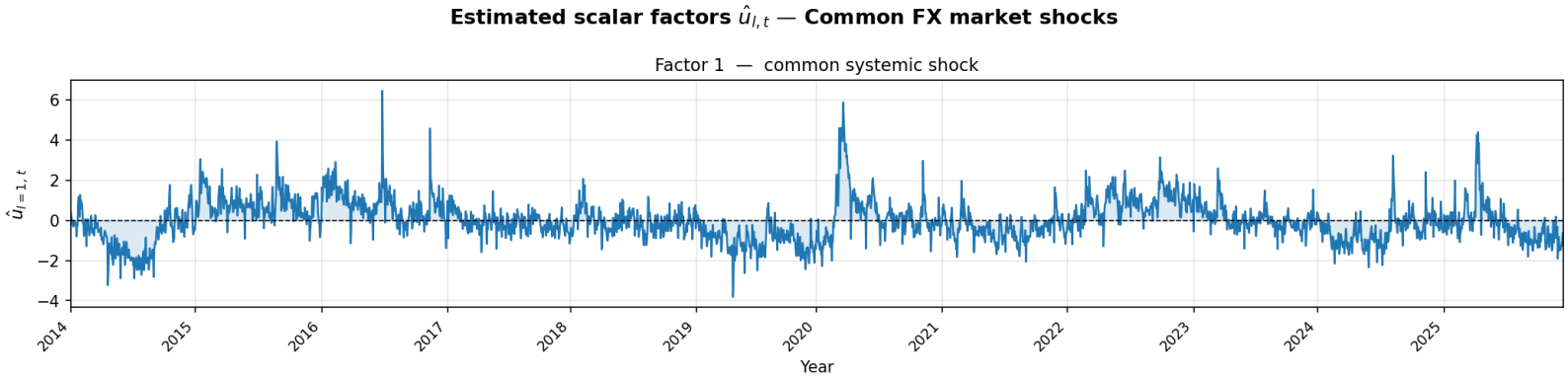


Figure: Estimated scalar factor score $\hat{u}_{1t}$, 2014-2025.

→ **Market-wide volatility index.** Largest spikes line up with systemic episodes:

- Brexit 2016 (≈ 6.5), COVID March 2020 (≈ 5.9)
- 2022 (Ukraine war + inflation), 2025 (US trade turmoil)

Strong clustering: high-volatility days arrive in groups.

FFM estimation - Results

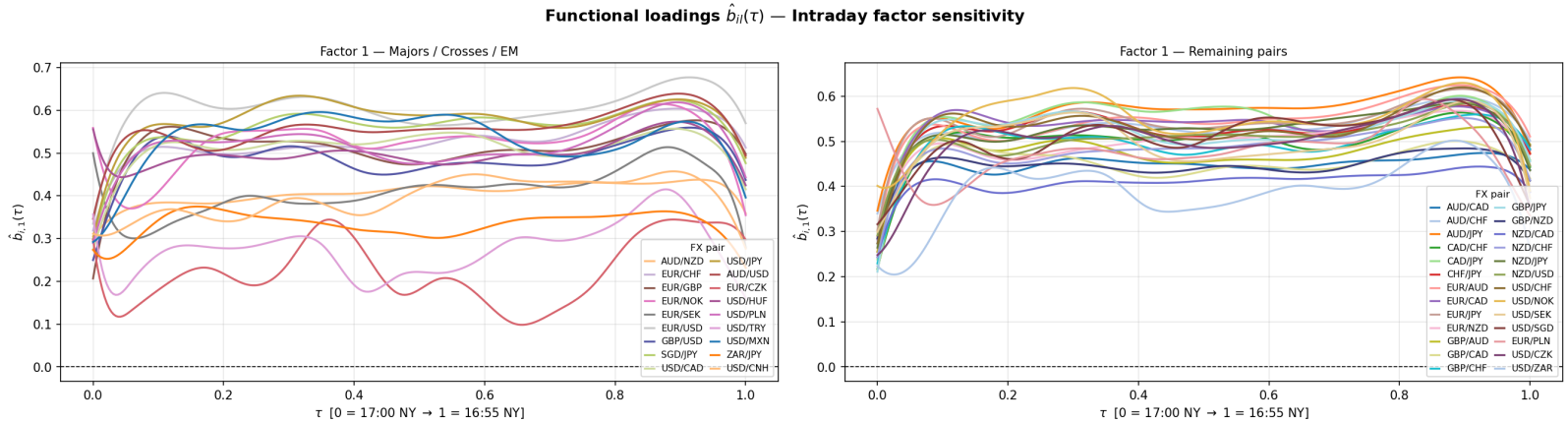


Figure: Functional loadings $\hat{b}_{i1}(\tau)$ for the 44 pairs.

- All loadings positive at every $\tau \rightarrow$ factor shifts every curve the same direction
- Mostly flat, with session-tied bumps: rise at the open, fall at the US close
- Similar amplitude across most pairs (comparable sensitivity)
- EM pairs (EUR/CZK, USD/TRY, ZAR/JPY) lower $\rightarrow$ co-move only weakly with the factor

Goodness of fit

$$R^2_{it} = \frac{\|\hat{\chi}_{it}\|^2_{L^2}}{\|\tilde{y}_{it}\|^2_{L^2}}$$

- Common component $= \mathbf{28.8\%}$ of avg. curve variance
- Brexit day: cross-sectional $R^2 = \mathbf{81.6\%}$
- Moderate on average, **dominant under global stress** (vertical bands)
- Pair-specific moves $\rightarrow R^2$ collapses; EM pairs persistently low

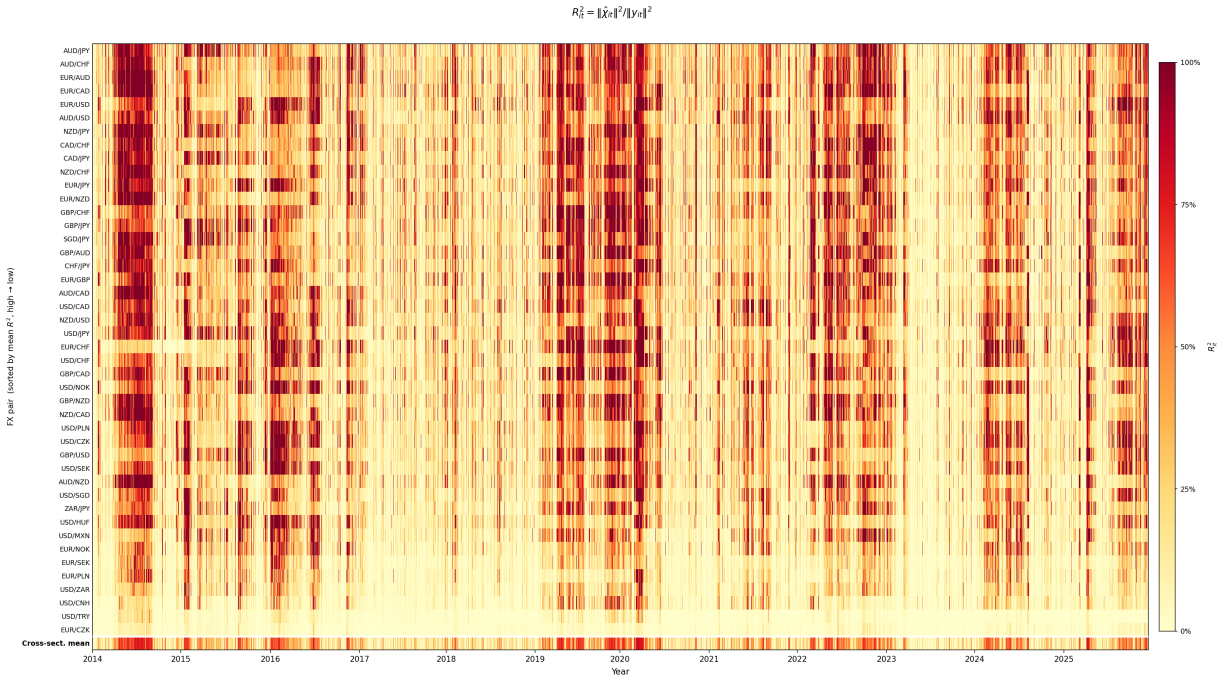


Figure: Per-pair daily R^2_{it}, 2014–2025.

Decomposition - Systematic & Idiosyncratic

Calm day

- Fitted curve tracks closely, $\|\hat{\xi}\|$ small ($\approx$ 5–6)

COVID (systemic)

- Common lifts the curve, residual moderate (10.2)

SNB 2015 (idiosyncratic)

- Curve jumps, fit barely moves $\rightarrow$ huge residual (63.2)

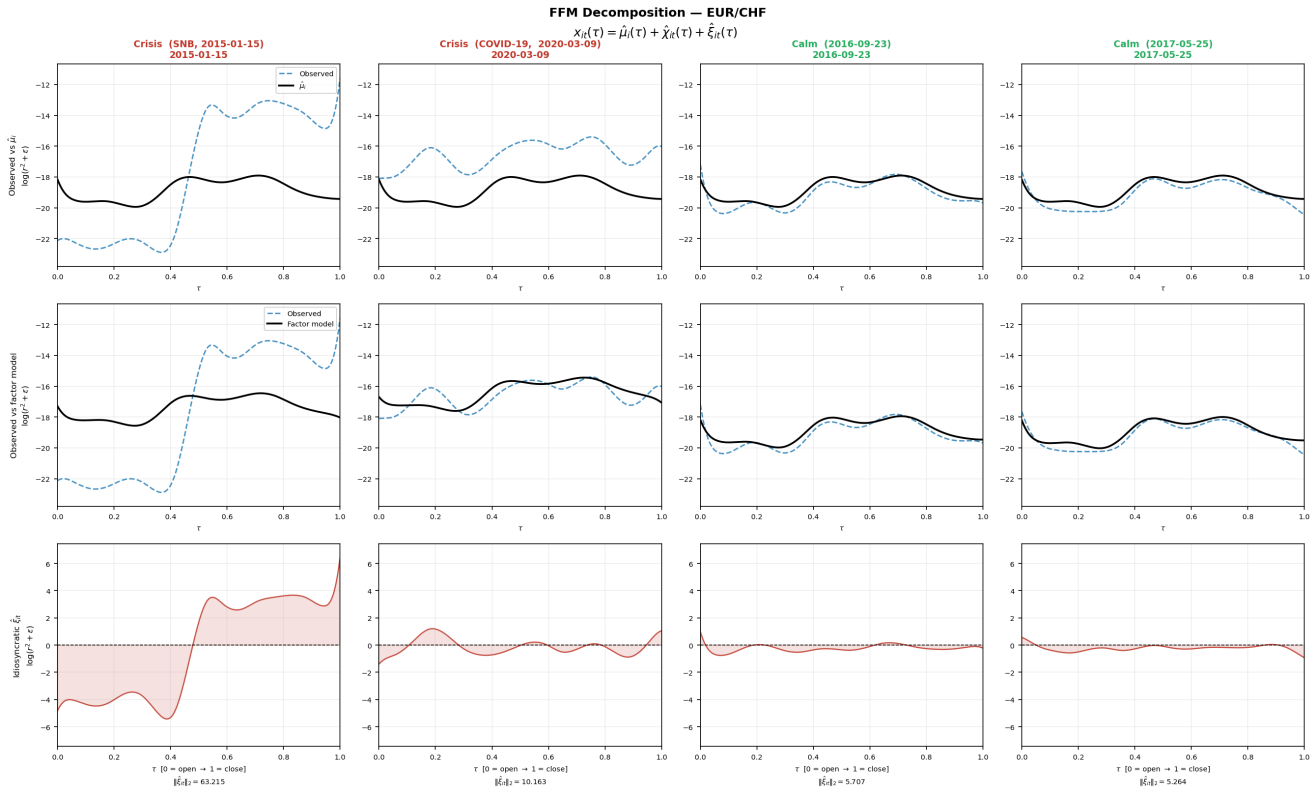


Figure: FFM decomposition for EUR/CHF: crisis vs. calm days.

Evolution of $\hat{r}$

- Full sample $\hat{r} = 1$; sub-period 2017–2025 $\hat{r} = 2$ (first sign)
- 6-month rolling windows, monthly step: ER/GR mostly 1, never above 3
- COVID: leading Gram eigenvalue explodes, ratio $\hat{\lambda}_1/\hat{\lambda}_2$ spikes

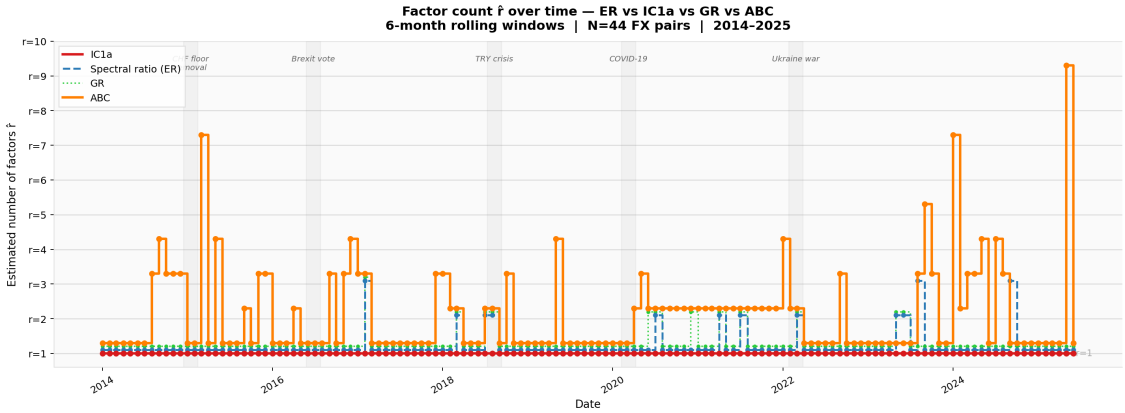


Figure: Rolling estimate $\hat{r}_w$ over 2014-2025.

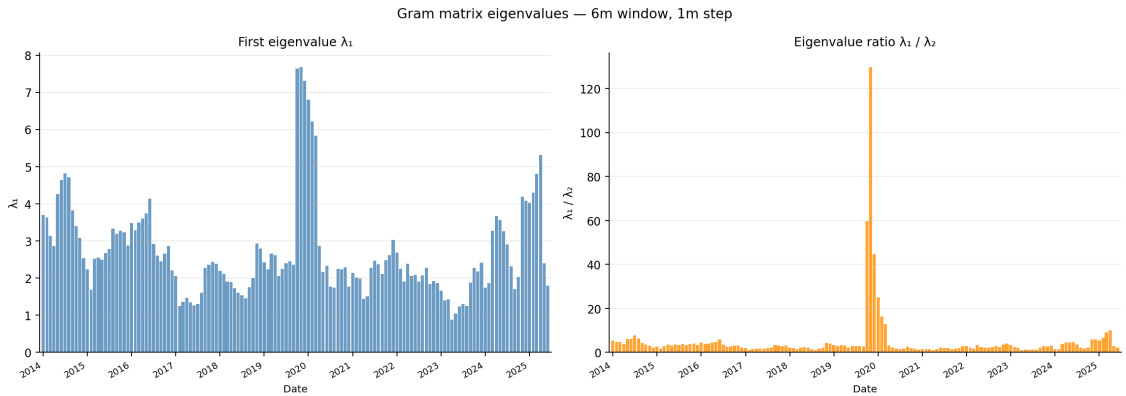


Figure: Rolling leading Gram eigenvalue and ratio.

Geometric Analysis - Grassmann distance

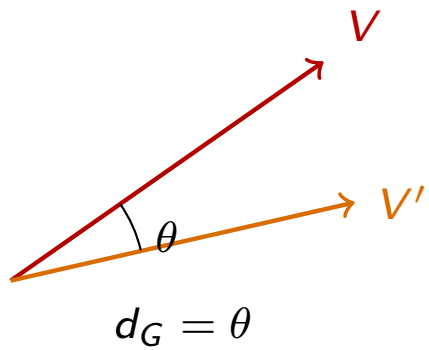
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Why Grassmann Distance ?

- Loadings identified only up to rotation $\rightarrow$ compare the *subspace*, not individual $\hat{b}$
- Two shock types: *level* ($\hat{\Gamma} \approx c\Gamma$, eigenfunctions fixed) vs. *structural* (rotation)
- A test on the mean misses a pure rotation
- Davis-Kahan: $|\sin\theta| \leq \dfrac{\|\Gamma - \hat{\Gamma}\|}{\delta}$ separates rotation from rescaling

Definition

- $\mathcal{G}(r, n)$: r-dim subspaces of $\mathbb{R}^n$; principal angles θ_k
- With $\hat{r} = 1$: $d_G = \arccos|\langle V, V' \rangle| \in [0, \frac{\pi}{2}]$
- Invariant to rotation, blind to level



Drift away from the 2014 baseline

- Distance to Jan–Jun 2014 window; range $[0.4, 1.53]$, mean ≈ 0.93
- Steady upward drift up to COVID $\rightarrow$ subspace progressively rotates
- **COVID**: distance drops to its minimum $\rightarrow$ radial shock, no rotation
- Distance and volatility level move apart (largest spike $\leftrightarrow$ smallest distance)

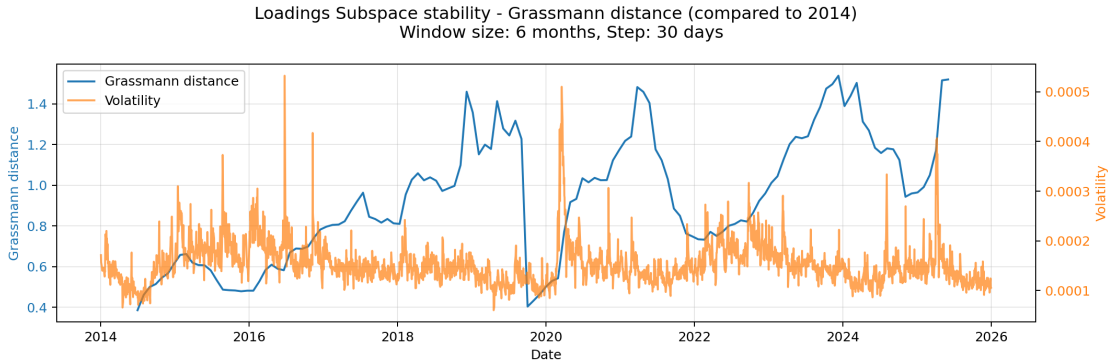


Figure: Fixed-baseline $d_G^{(w)}$ (blue) and daily mean volatility (orange).

Block structure across windows

- Dark diagonal → consecutive windows barely rotate
- 2014–2018 and 2020–2023 form two internally-stable blocks
- 2019 a bright cross; COVID sits near the calm 2014–2017 windows
- Per-pair: EUR/CZK, EUR/PLN drift most; shared moves 2017–18, 2021, 2024

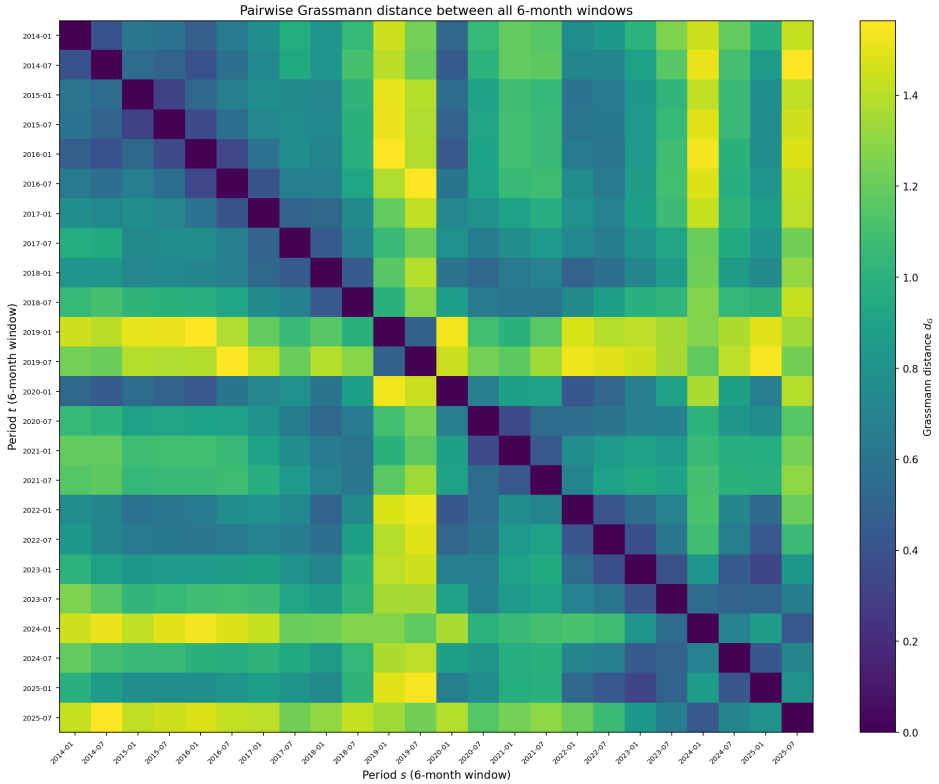


Figure: Pairwise d_G between all 6-month windows.

Simulation - Null distribution

H_0: every window shares the 2014 leading eigenspace

- Simulate $Z \sim \mathcal{N}(0, \Sigma_{ref})$; two samples $S = 132$; $B = 1000$
- Estimate each leading direction, compute d_G
- Null: $d_G \in [0.12, 0.33]$ (95% < 0.24)
- **Observed:** $d_G \in [0.40, 1.53]$ $\rightarrow$ clear separation

$\Rightarrow$ rotation $>$ sampling variability

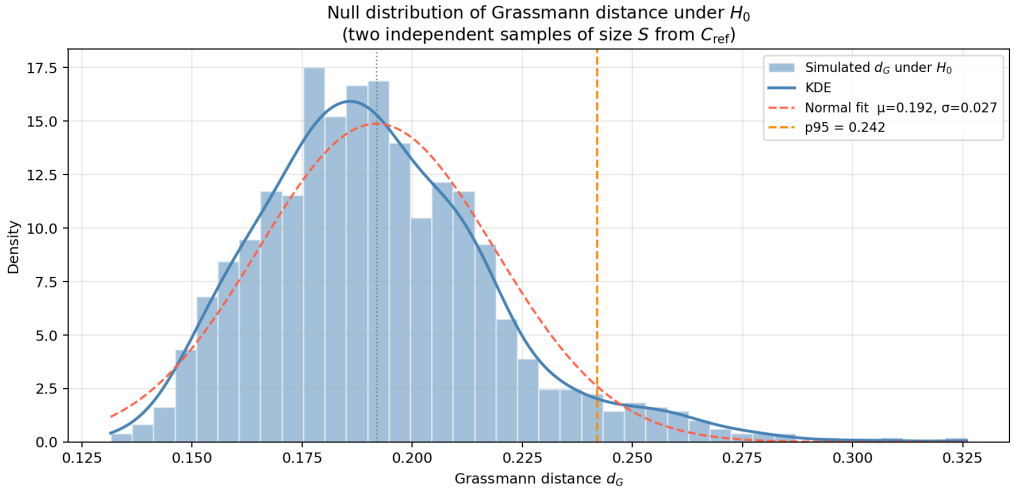


Figure: Simulated null distribution of d_G under H_0.

CUSUM + Bai-Perron

CUSUM (single mean shift)

- Not rejected → series rises, drops at COVID, rises again → moves cancel

Bai-Perron (multiple regimes)

- BIC selects 7 breaks → 8 regimes
- COVID regime has a *lower* mean than neighbours → level event

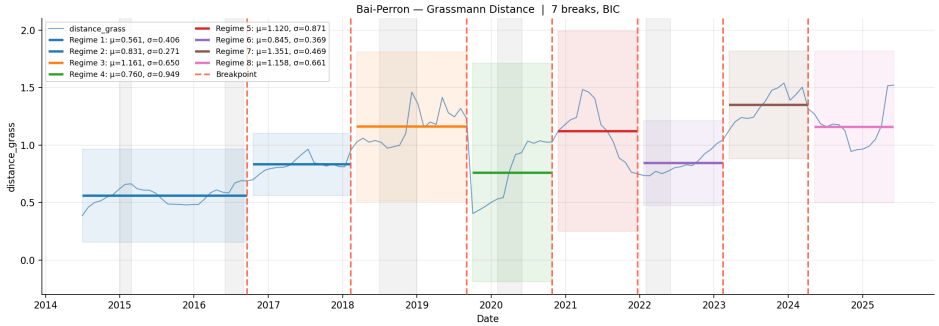


Figure: Bai-Perron segmentation of $d_G^{(w)}$ (7 breaks, BIC).

Dette & Kutta - Break Structure detection

Tests for a change point in the eigenvalues / eigenfunctions of the covariance operator:

$$H_0^{DK}: \|v_j^{(1)} - v_j^{(2)}\|^2 \leq \Delta_v \quad \text{vs.} \quad H_1^{DK}: \|v_j^{(1)} - v_j^{(2)}\|^2 > \Delta_v, \qquad \Delta = 0.$$

- Joint break $\hat{\theta} = 0.678$ (2022-02-17); eigenvalue break 2016-07-15; eigenfunction break 2018-11-16
- At $\Delta = 0$ (most demanding): **no break significant**, and the three dates do not coincide
- Complementary to Grassmann: descriptive rotation (Grassmann) vs. formal verdict (Dette–Kutta)

Component	**Break**	$\hat{E}/\hat{D}$	**stat**	**Reject ($\Delta = 0$)**
Eigenvalue	2022-02-17	2.63	0.97	No
Eigenfunction	2022-02-17	0.077	5.35	No
Eigenfunction	2018-11-16	0.076	6.26	No

($q = 9.90$)

Level vs Structure

Level shock

- Rescales the operator $\hat{\Gamma} \approx c\Gamma \rightarrow$ eigenfunctions fixed
- Absorbed by the score / loading amplitude
- **COVID**: eigenvalue explodes, but d_G drops; returns near 2014 at peak vol

Structural shock

- Rotates the eigenfunctions (Davis–Kahan)
- Slow drift 2017+, 2019 episode, renewed drift 2023–24
- Shows only in the geometry, not the volatility level

Answer: the subspace is *not* stable (distances $\gg$ null), but the instability is *gradual and multi-regime* $\rightarrow$ stable within regimes, reorienting across them; not a single datable break.

Out-of-sample Performance

Target: the intraday curve on 2026-01-02 (one step ahead). With $\hat{r} = 1$, $\hat{\chi}_{it}(\tau) = \hat{b}_{i1}(\tau)\,\hat{u}_{1t}$.

1. Score
   - Forecast the scalar $\hat{u}_{1,T+1}$ (one number)
2. Idiosyncratic
   - Forecast $\hat{\xi}_{i,T+1}(\tau)$ for each pair
3. Recombine
   - $\hat{x} = \hat{\mu}_i + \sigma(\hat{\chi} + \hat{\xi})$ on the original scale

Model

Component models

- **Score**: stationary (ADF, KPSS) but persistent $\rightarrow$ AR(5), coeffs ≈ 0.92
- **Idiosyncratic**: not white noise (functional Ljung-Box rejects all 44 pairs)
- Three specs: Random Walk, FAR, HAR (via FPCA)

Results

- Idiosyncratic side drives gains: HAR $\approx -20\%$ MAFE, -41% MSFE
- Factor side barely matters: AR $\approx$ RW (score nearly a random walk)
- HAR best; CHF crosses the only laggards

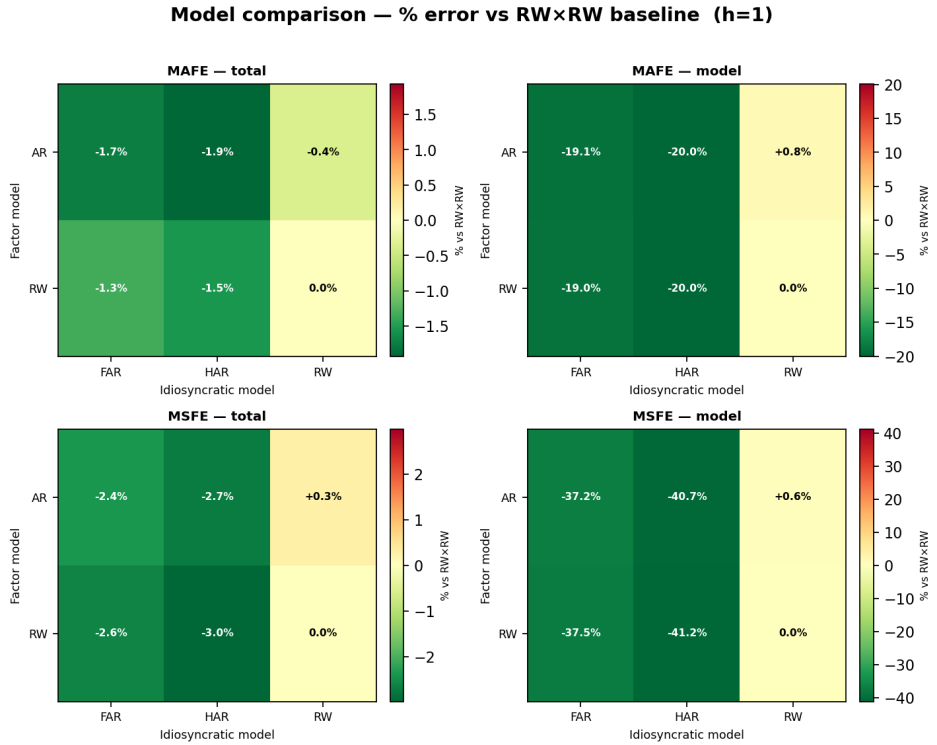


Figure: Error reduction vs. RW$\times$RW baseline (model target).

Loading instability & Forecast degradation

- Freeze loadings at 2014 vs. use each window's own loadings
- Degradation $\Delta_w = \mathrm{MSFE}^{(ref)} - \mathrm{MSFE}^{(w)}$ grows with distance from reference
- Δ_w positively correlated with d_G ($\rho \approx 0.26$)
- **COVID**: $\Delta_w \approx 0$ - level shock, loading choice irrelevant

$\Rightarrow$ rotation carries a real, growing forecast cost

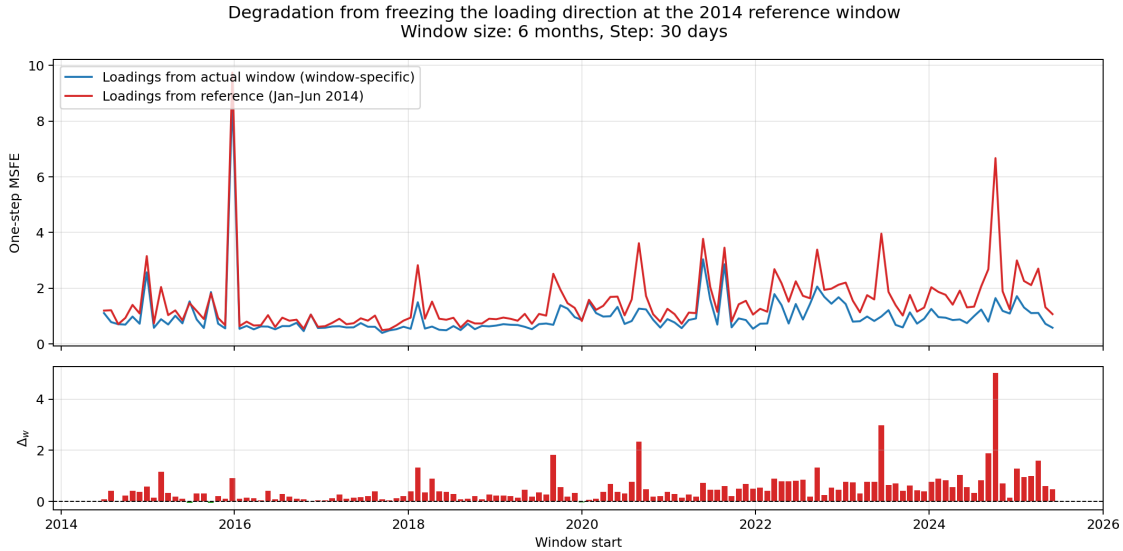


Figure: One-step MSFE (own vs. 2014 loadings) and Δ_w.

Conclusions

- **One interpretable factor** $\rightarrow$ market-wide volatility index, 28.8% of intraday curve variance, far more under stress; G10 loads strongly, EM weakly
- **Main contribution**: geometric & statistical analysis of loading-subspace stability $\rightarrow$ it rotates more than sampling noise allows
- **Gradual, not a break**: several moderate regimes; level shocks (COVID) vs. structural drift (2017+, 2019, 2023–24) separated
- **It costs forecasting**: freezing rotated loadings degrades the one-step forecast, and the cost grows with the Grassmann distance

Limitations & Extensions

Limitations

- All pairs weighted equally (ignores volume / G10 dominance)
- No external covariates to explain *when* rotation happens
- Cross-section not independent → shared legs, triangular arbitrage
- Forecast comparison on a single date (illustrative, not a backtest)

Extension

- A formal test for the *gradual* instability itself
- Break tests have little power vs. a slow, monotone rotation

Thank you

Questions & discussion

Supervisors: Prof. Victor Panaretos Prof. Shahin Tavakoli

Application to the Raw Data

